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(54) **X-RAY DETECTOR FOR ELECTRON MICROSCOPE**

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(60) Provisional application No. 61/122,295, filed on Dec. 12, 2008.

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H01J 37/24 (2006.01)
G01N 23/04 (2006.01)

(52) **U.S. Cl.** **250/307**; 250/251; 250/305; 250/306; 250/311; 378/46; 378/70; 378/71

(58) **Field of Classification Search** 250/251, 250/305, 306, 307, 311; 378/46, 70, 71
See application file for complete search history.

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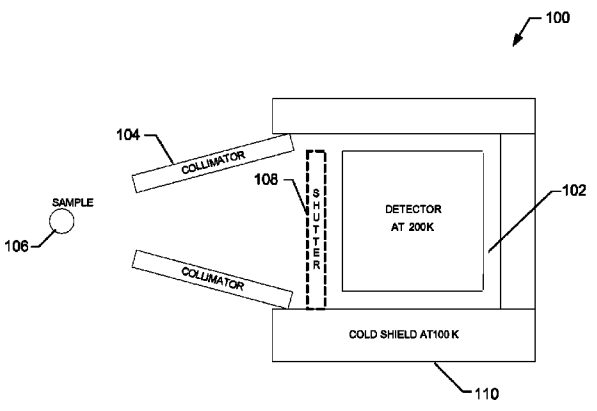
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(57) **ABSTRACT**

Multiple detectors arranged in a ring within a specimen chamber provide a large solid angle of collection. The detectors preferably include a shutter and a cold shield that reduce ice formation on the detector. By providing detectors surrounding the sample, a large solid angle is provided for improved detection and x-rays are detected regardless of the direction of sample tilt.

21 Claims, 5 Drawing Sheets



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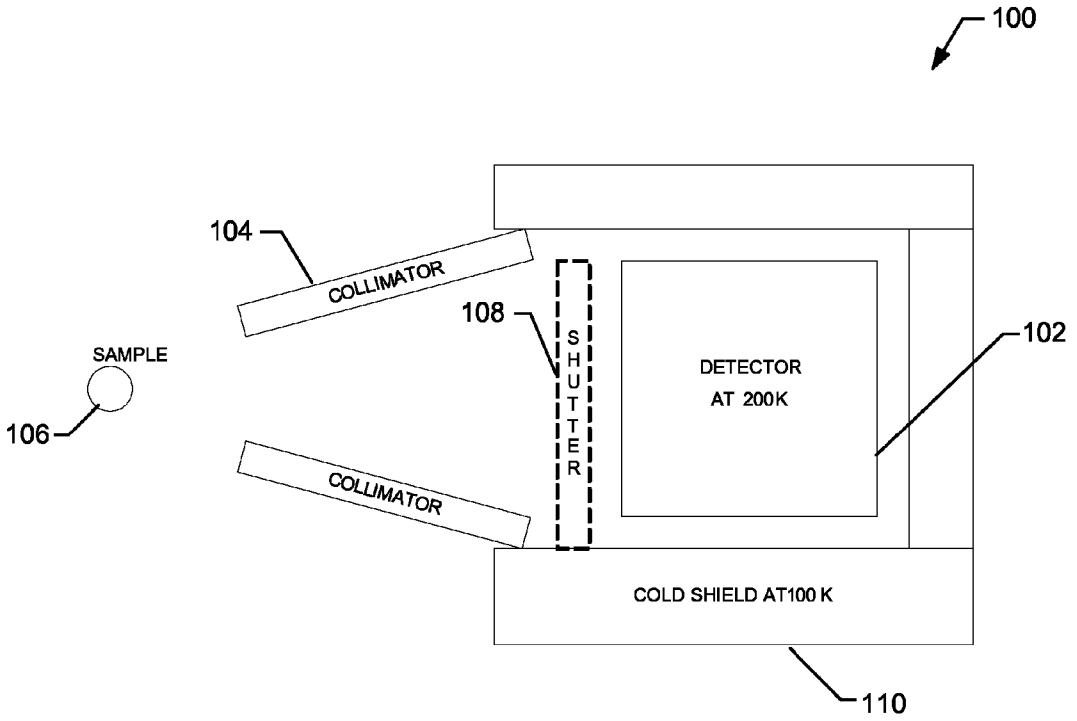

100
104
COLLIMATOR
108
SHUTTER
102
SAMPLE
106
DETECTOR
AT 200K
COLLIMATOR
COLD SHIELD AT100 K
110

FIG. 1

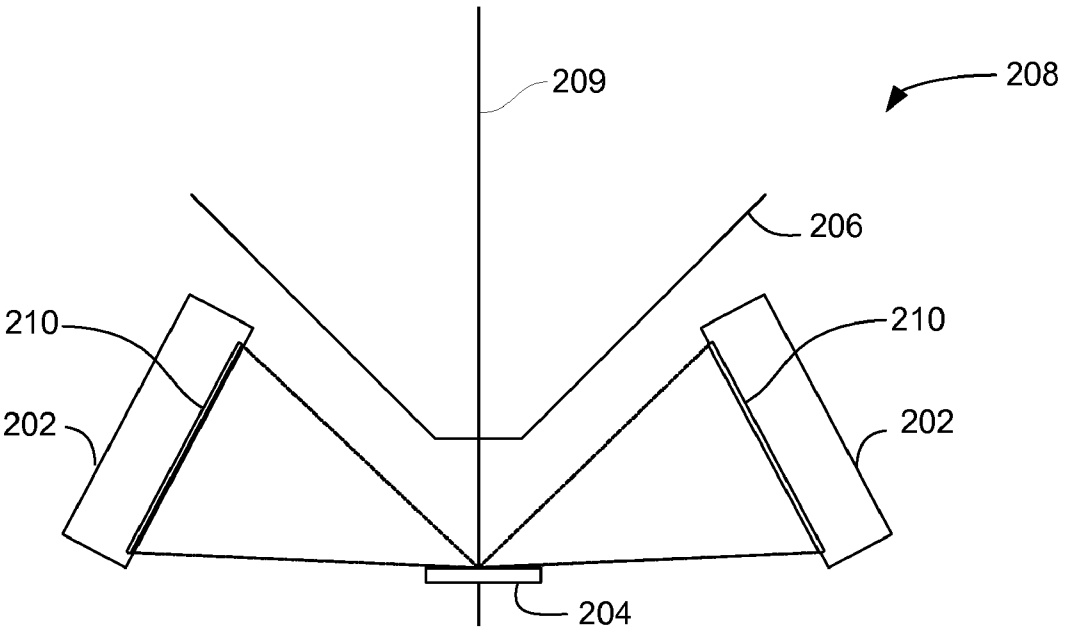


FIG. 2

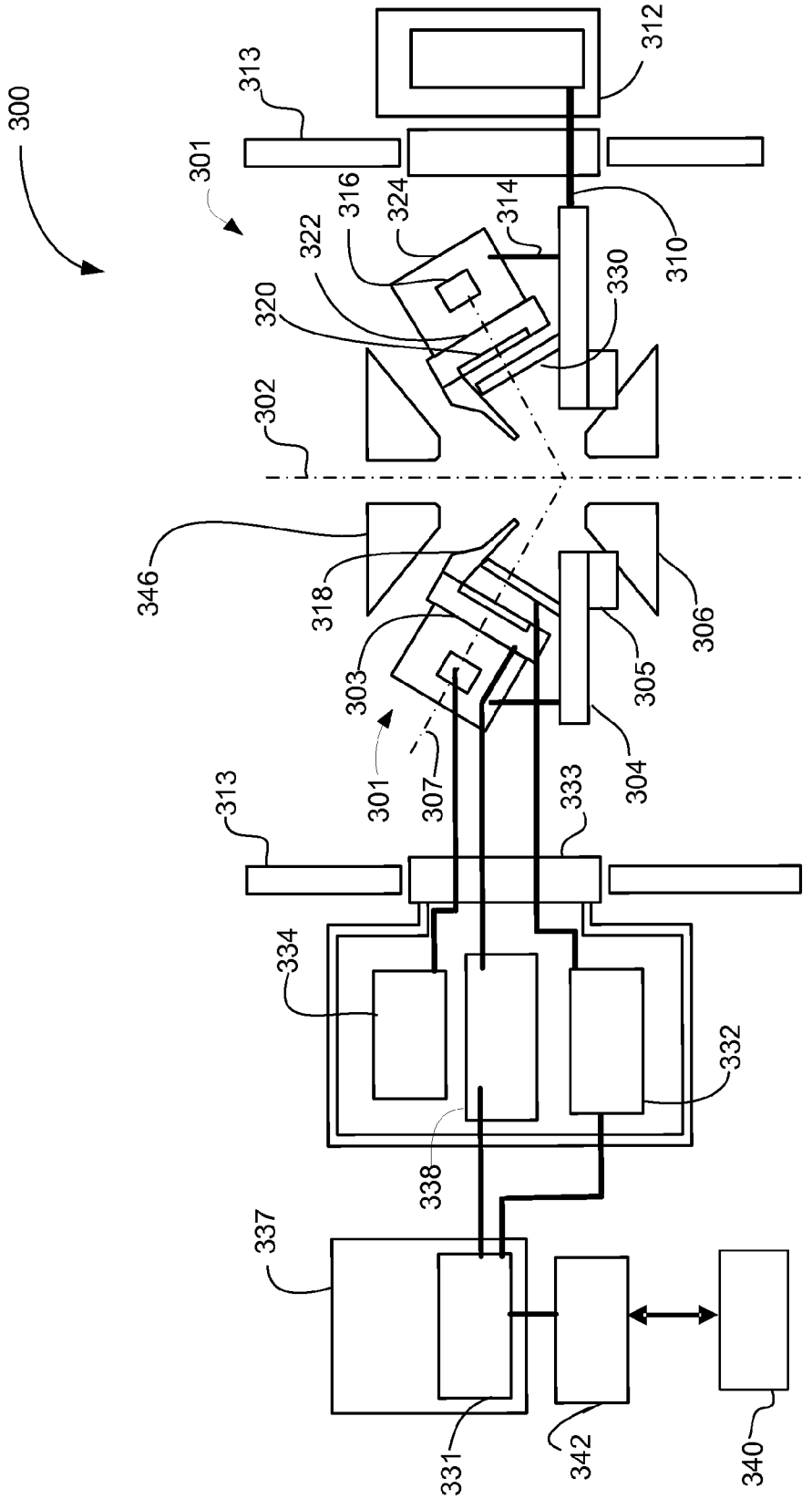

300
301
302
303
304
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342
346

FIG. 3

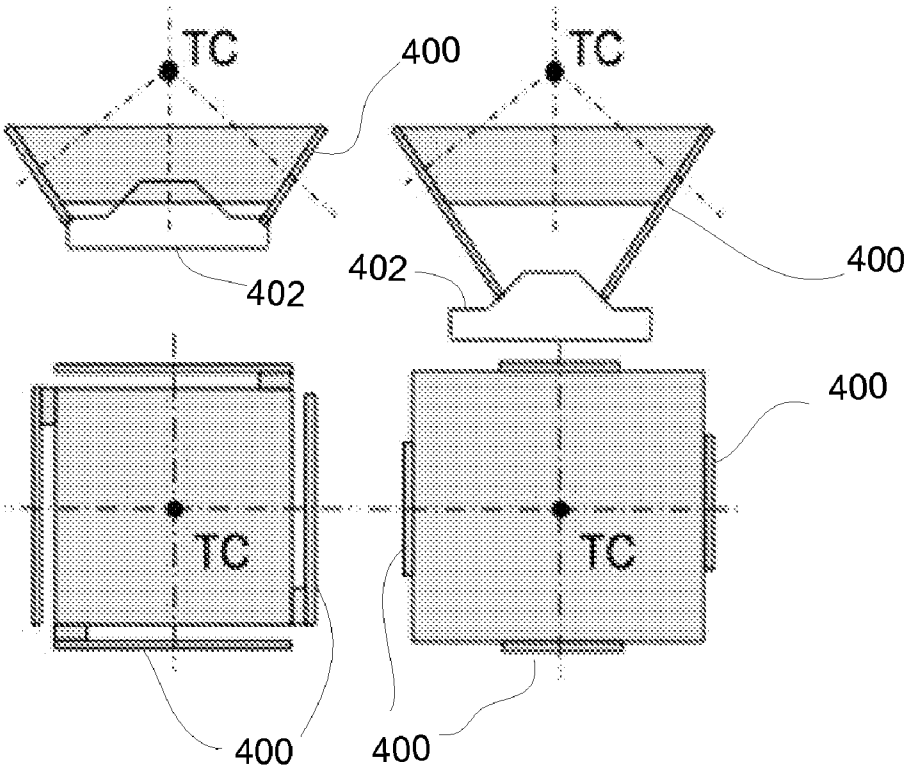


FIG. 4A FIG. 4B

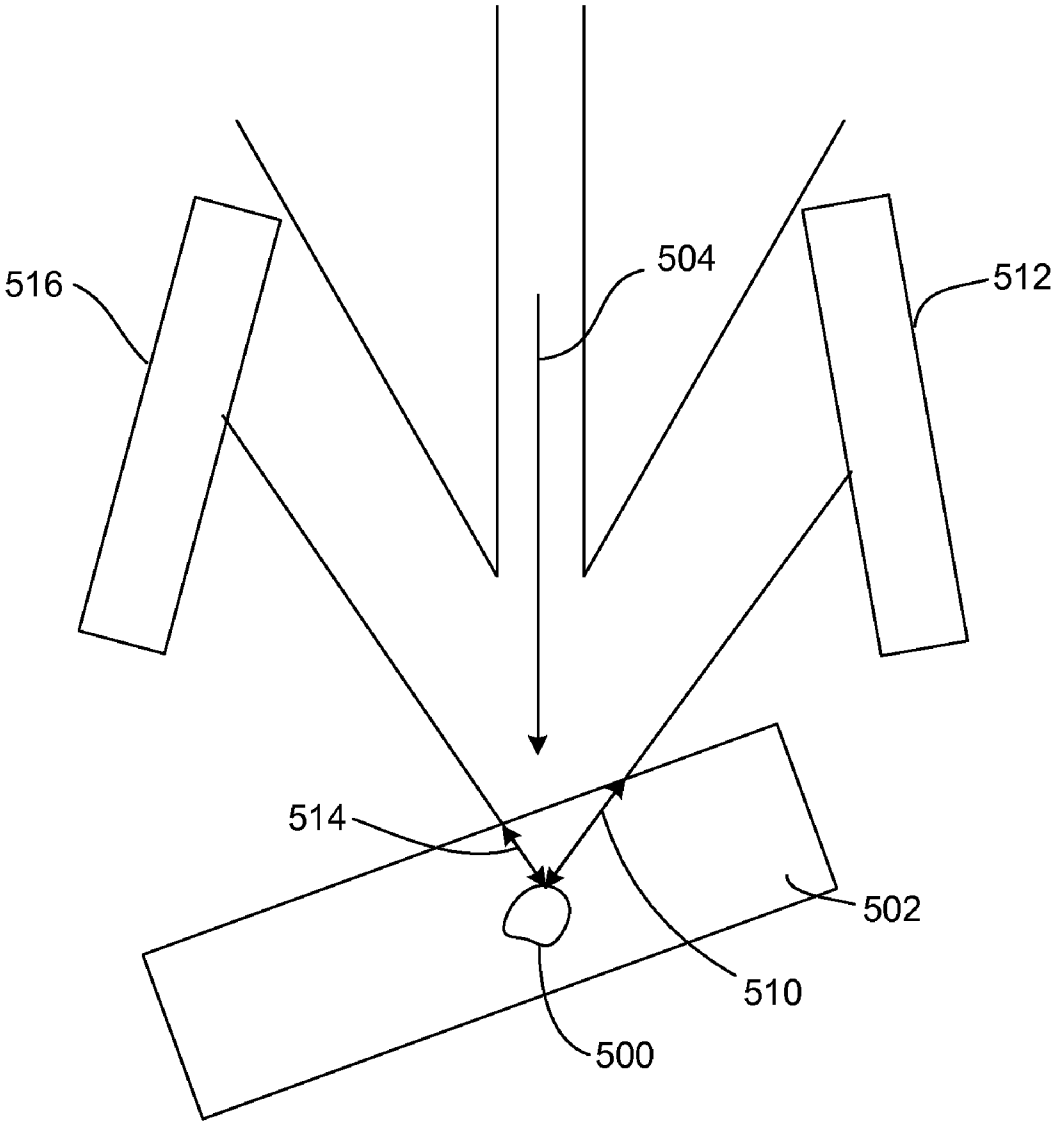


FIG. 5

X-RAY DETECTOR FOR ELECTRON MICROSCOPE

This application is a continuation of U.S. patent application Ser. No. 12/494,227, filed Jun. 29, 2009, and issued as U.S. Pat. No. 8,080,791, which claims priority from U.S. Provisional Pat. App. No. 61/122,295, filed Dec. 12, 2008, both of which are hereby incorporated by reference.

TECHNICAL FIELD OF THE INVENTION

The present invention relates to x-ray detectors, and in particular, to an x-ray detector for an electron microscope.

BACKGROUND OF THE INVENTION

Electron probe microanalyzers and electron microscopes having an attached x-ray spectrometer are used to determine the composition of microscopic or nanoscopic regions of a surface. The detectors determine the energy or wavelengths of x-rays emitted from the sample and infer the composition of material under the electron beam from the energy or wavelength of the x-rays. Detectors that use a crystal to disperse and analyze x-rays of different wavelengths are referred to as wavelength dispersive spectrometers (WDS) and detectors that measure the energy of incoming x-rays are referred to as energy dispersive spectrometers (EDS). While a WDS can provide better resolution and faster counting for a particular wavelength of x-ray, an EDS is better adapted to measuring x-rays of different energies from multiple elements.

Two types of semiconductor energy dispersive x-ray detectors are commonly used in electron microscopy: lithium-drifted silicon detectors "Si(Li)" and silicon drift detectors "SDD". Si(Li) detectors typically require cooling to liquid nitrogen temperatures and normally have a standardized active detection of area of 10, 30 and 50 mm^2. SDDs can operate at a higher temperature and can provide better resolution at high count rates. To avoid ice formation and contamination on the detector, as well as damage from backscattered electrons, a window of a light element such as berrylium is often attached in front of the detector to stop the electrons. A magnetic field can also be used near the detector entrance to divert electrons away from the detector. A collimator is often used in front of the detector to reduce x-rays from sources other than the sample from entering the detector. Some detectors, such as the one described in U.S. Pat. No. 5,569,925 to Quinn et al., include a shutter in front of the detector. When the electron microscope is operated under conditions that would generate high energy x-rays and electrons that could damage the detector, the shutter can be closed to protect the crystal.

Ice formation is also reduced by providing a colder surface near the detector. For example, in the system described in U.S. Pat. No. 5,274,237 to Gallagher et al. for a "Deicing Device for Cryogenically Cooled Radiation Detector," the heat generated by the detector circuitry maintains the detector a few degrees warmer than the collimator surface so that moisture sublimes from the detector surface onto the collimator surface. The heat generated by the circuitry provides a temperature difference of only about five degrees, which may not be adequate to maintain an ice-free surface on the detector. U.S. Pat. No. 4,931,650 to Lowe et al. for "X-ray Detectors" describes periodically heating the detector above its operating temperature while maintaining a heat sink at operating temperature. Periodically heating the detector above its operating temperature does not stop the build-up of ice during operation and requires periodic interruption of the system operation to remove the ice.

For greatest sensitivity, the detector should cover a large solid angle from the sample to collect as many of the emitted x-rays as possible. To increase the solid angle, the detector can provide a larger active surface area, or be placed closer to the sample. In a transmission electron microscope, the pole pieces and sample holder take up most of the space around the sample and it can be difficult to position X-ray detectors close to the sample to increase the solid angle. U.S. Pat. No. 4,910,399 to Taira et al. teaches a configuration that puts a detector closer to the sample and allows the detector to subtend a larger solid angle. Another configuration is shown in Kotula et al., "Results from four-channel Si-drift detectors on an SEM: Conventional and annular geometries," *Microscopy and Microanalysis,* 14 Suppl 2, p. 116-17 (2008). Kotula et al. describe a four-segment detector, with each segment being kidney-shaped and having an active area of about 15 mm^2. The detector is positioned above the sample below the pole piece of an SEM, with the four segments distributed in a ring that is coaxial with the electron beam. This configuration is not normally possible in a high-resolution TEM.

SUMMARY OF THE INVENTION

An object of the invention, therefore, is to provide an x-ray detector having improved detection capabilities.

A preferred embodiment uses multiple detectors arranged in a ring within a specimen chamber to provide a large solid angle of collection. The detectors preferably include a shutter and a cold shield that reduce ice formation on the detector. By providing detectors surrounding the sample, a large solid angle is provided for improved detection and x-rays are detected regardless of the direction of sample tilt.

The foregoing has outlined rather broadly the features and technical advantages of the present invention in order that the detailed description of the invention that follows may be better understood. Additional features and advantages of the invention will be described hereinafter. It should be appreciated by those skilled in the art that the conception and specific embodiments disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present invention. It should also be realized by those skilled in the art that such equivalent constructions do not depart from the spirit and scope of the invention as set forth in the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more thorough understanding of the present invention, and advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

FIG. **1** shows schematically a preferred detector of the present invention;

FIG. **2** shows the arrangement of detectors around a sample;

FIG. **3** shows the mechanical layout of a preferred detector embodiment;

FIGS. **4**A and **4**B show structures for mounting the detectors in which the target point of the detectors remains relatively constant as the temperature changes; and

FIG. **5** shows how the difference in x-ray path length can be used in x-ray tomography without requiring a series of tilt images.

The accompanying drawings are not intended to be drawn to scale. In the drawings, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labeled in every drawing.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

A preferred embodiment uses multiple detector assemblies arranged in a ring within a specimen chamber to provide a large solid angle of collection. FIG. **1** shows schematically a preferred embodiment of one of the multiple detector assemblies **100**. Each detector assembly **100** includes a SDD detector **102** having an active area of preferably greater than 10 mm^2, more preferably greater than 20 mm^2, and even more preferably about 30 mm^2. In some embodiments, each detector includes an active area of between 50 mm^2 and 100 mm^2. The detector of FIG. **1** includes a collimator **104** that prevents stray x-rays from entering the detector, ensuring that the signal from the detector corresponds to the x-rays emitted from a sample **106**. A mechanical shutter **108**, shown in the closed position, prevents electrons and low-energy x-rays from damaging the detector when it is not in use.

The SDD detector **102** is cooled to about 200 K using liquid nitrogen and is surrounded by a cold shield **110** maintained at about 100 K. In other embodiments, the detectors are maintained at temperatures of between about −60° C. and about −80° C. for optimum detector performance. However, it is possible to operate the SDD at higher temperatures, up to and including room temperature. Harmful background gases in an electron microscope are mostly water vapor and hydrocarbons, such as, for example, oils. Ice and hydrocarbons tend to condense onto the detectors, absorbing some incident x-rays and reducing the collection efficiency. Maintaining the detector cold shield at a temperature that is significantly colder than the detector and operating in an ultra high vacuum prevents the build-up of ice on the detector. The temperature difference between the detector and the cold shield is preferably greater than 10° C., more preferably greater than 25° C., even more preferably greater than 50° C., and most preferably equal to or greater than about 100° C. The shutter, which is maintained at about the same temperature as the cold shield, also protects the detector from ice formation. By reducing ice formation on the detector, a preferred detector does not require a window. Eliminating the window improves the detection efficiency for low-energy x-rays. Some embodiments, however, still use windows of a material, such as beryllium or thin polymer foils, which minimize x-ray absorption. Collimators **104** preferably are in thermal contact with cold shield **110** to provide additional cold shielding.

FIG. **2** shows a preferred arrangement of detectors **202** positioned around a sample **204** below, and thermally isolated from, an upper pole piece **206** of a TEM **208** having an optical axis **209**. A preferred embodiment includes four detectors arranged around the sample, although only two are visible in FIG. **2**. The two detectors **202** are positioned on opposite sides of the eucentric sample stage tilt axis, which is normal to the plane of the drawing. An active area **210** of each detector preferably subtends a solid angle of about ¼ steradian or greater to provide a total detector solid angle of about a steradian or greater. Each detector preferably detects x-rays at angles of less than 50 degrees from the sample surface, more preferably at angles of less than 35 degrees from the sample surface, and more preferably at angles between 5 and 35 degrees from the sample surface. The low take-off angle reduces damage from backscattered electrons, while maintaining bremsstrahlung radiation, which also gives rise to background in the spectra, but at a reasonably low level. The low take-off angle detectors also require no space between the pole pieces, thereby not requiring a degradation of the TEM or STEM resolution and sample tilt range. When the term "take-off angle" is used to describe a location or orientation of a detector, it is assumed that the sample surface is horizontal.

FIG. **3** shows schematically a TEM **300** including four x-ray detectors assemblies **301** (only two shown) positioned around a TEM optical axis **302**. Each x-ray detector assembly **301** includes an X-ray detector **303**, which is preferably an SDD detector, which is known in the art. X-ray detector assemblies **301** are supported by a support ring **304** which is in turn supported by a lower pole piece **306** with a thermal insulator **305** thermally isolating support ring **304** from lower pole piece **306**. The optical axis **307** of each detector assembly **301** preferably intersects the optical axis **302** of the TEM at the sample surface. Thermal conductor **310**, preferably a copper braid or rod, connects support ring **304** with a cold source, such as a Dewar flask **312** of liquid nitrogen maintained outside of vacuum chamber walls **313**. A support **314** provides support to each detector assembly **301** from support ring **304** and provides a resistive thermal path from support ring **304** to detectors **303**, that is, support **314** provides a thermal path between support ring **304** and detector **303**, but restricts the thermal flow. A resistive heater **316** maintains the temperature of each detector **303** significantly greater than the temperature of support ring **304**. Detectors **303** are preferably maintained at a temperature difference of greater than 20° C., greater than 50° C., or approximately equal to or greater than 100° C., relative to support ring **304**. A collimator is formed by an upper collimator **318** and a portion of support ring **304**. The use of an SDD detector, which can operate efficiently at a temperature significantly higher than the optimum operating temperature for a Si(Li) detector, allows the detector to be operated at a temperature significantly above the temperature of liquid nitrogen. This facilitates the use of a single liquid nitrogen cold source to cool both the detector and a cold shield, while allowing the cold shield to be maintained at a much lower temperature than the detector, which improves protection of the detector from contamination by condensates.

Each detector **303** includes an active area **320** positioned on a ceramic substrate **322** supported by a metal base **324**. Four shutters **330** protect the four detectors **303** from ice formation and from high energy electrons when closed. During operation, the shutters are moved to an open position to allow x-rays to reach the active area **320**. The shutters are in thermal contact with support ring **304**, which keeps them at about the same temperature as support ring **304** and at a temperature significantly lower than the temperature of detector **303**. Support ring **304**, shutters **330**, and preferably upper collimator **318**, function as a cold shield, causing moisture to sublime from the detector active area **320** onto the support ring **304**, shutters **330**, and collimator **318**. The shutters are controlled by a shutter controller **331** through a shutter activator **332** which connects to the shutters through a mechanical linkage via a vacuum feed-through **333** from outside the vacuum chamber walls **313**. In one embodiment, each shutter activator activates two shutters. TEM **300** is controlled by a user **340** through the TEM Personal computer **342**.

A heater controller **334** controls the current to heater **316** to maintain a desired temperature of detector active area **320**. A pre-amp **338** receive a signal from the detector **303**. The signal is processed by a pulse processer **337** to determine the

number of x-rays counted and the energy of each x-ray. Pulse processing techniques are well known in the EDS art.

Prior art detectors were typically attached to the vacuum chamber wall, more than 10 cm away from the sample, which made it difficult to keep the detectors aligned with the sample. When a detector is even 0.2 mm out of position, it is no longer "looking" at the point at which the x-rays are generated, and therefore picks up unwanted signals. In preferred embodiments of the present invention, the detector assembly is attached within the TEM lens assembly, preferably to the lower pole piece. Because the detector is typically cooled to cryogenic temperature, it would have been considered undesirable to attach the detector assembly to the pole pieces, because it could lead to thermal instability. Applicants have found, however, that insulation **305** between the detector assembly and the pole pieces reduces thermal instabilities. The detector can also be secured to or against an upper pole piece **346** to provide additional alignment and stability. Moreover, the electric field from the detector, particularly a windowless detector, affects the electrons in the primary beam. Applicants have found that any deflection in the primary electron beam is relatively small and that the improved accuracy of the detector positioning outweighs any disadvantages of the mounting. Mounting the detector assembly onto the lens provides improved mechanical accuracy relative to the specimen.

FIGS. **4**A and **4**B show two designs for supporting a detector support ring **402** that in turn supports at least two individual detectors assemblies. Each design is shown in a front view and a top view. Both designs support the detector support ring using flexible elements **400**, such as leaf springs, that are connected on the one end to the detector support ring and on the other end to a support, such as a portion of the lens. The detector support ring is preferably supported within the TEM lens, preferably on the lower pole piece (not shown). The flexible elements provide a floating thermal and vibration isolation platform on which to mount the detectors.

It is desirable that the detectors maintain their aim on the intersection of the optical axis and the sample surface as the detectors are cooled to operating temperature so that the detector can be aligned at room temperature and then stay aligned as it is cooled. In the design of FIG. **4**A, the thermal center, that is, the point on the sample being analyzed, preferably does not move as the temperature of the detector changes. That is, the detectors are always 'looking' at approximately the same point: the intersection of the sample with the axis of the microscope as the temperature changes. The symmetry of the four leaf springs provides a slight rotation about the microscope axis as the assembly is cooled to operating temperature, as well as a radial contraction. The material of the leaf springs and the construction of components are selected to provide thermal coefficients of expansion that provide thermal stability such that the thermal center of the detectors on the microscope axis changes by less than 10 μm when the assembly is cooled from room temperature to cryogenic temperatures, such as to 100 K for the support ring. The thermal expansion coefficients of the different materials used to support the detector support are matched to achieve a net near-zero displacement of the thermal center. The thermal gradient in the flexible elements during operation is preferably linear, while the thermal gradient in the detector support ring is minimal. The mass and the dimensions of the leaf springs raise the resonance frequency of the mount to minimize vibration. While leaf springs are described to float the detector mount, other types of mounts can also be used.

In the design of FIG. **4**A, a change in temperature causes expansion of the flexible elements, which causes a slight rotation about the point TC on the optical axis and a radial contraction. The rotation about the point TC is not detrimental since x-ray emission is isotropic in the azimuthal direction. In the design of FIG. **4**B, a change in temperature causes translation parallel to the optical axis, which is less preferred.

The support ring may include an opening that accepts the upper pole piece of a TEM and a thermal isolation material thermally isolates the cold components from the upper pole piece. Openings in the ring support can be provided for inserting the sample, an aperture, or other devices. By mounting the support ring on the lower pole piece, the positioning of the x-ray detectors relative to the sample is maintained more accurately than the positioning of prior art detectors, which are mounted onto the walls of the vacuum chamber.

The invention allows three-dimensional X-ray tomography and depth determination of features in the specimen without requiring a series of images at different tilt angles. In the prior art, X-ray tomography was performed by obtaining a series of images at different sample tilts. The images at the different tilts were analyzed by a computer to determine the three dimensional structure of the sample region. Because the present invention provides multiple detectors, it is possible to use the difference in signal strengths between the detectors to determine the depth of a feature or the three dimensional distribution of materials.

FIG. **5** shows a defect **500** in a sample **502** under an electron beam **604**. X-rays emitted from the defect travel through the sample a distance **510** before reaching detector **512**. X-rays emitted from the defect travel through the sample a distance **514** before reaching detector **516**. The x-rays are attenuated as they travel through the sample, and so the x-ray signal at detector **512** will be different from the x-ray signal at detector **516**. The X and Y coordinates of the point at which the x-rays are generated is known from the known position of the scanned image. The Z position of the point from which the x-rays are generated can then be determined from the differences in signal strength at the different detectors, based on the different path lengths through the sample material.

Another use of the multiple detectors is differential x-ray detection, i.e. subtracting the signal from the 4 detectors in some combinations like (A+B)−(C+D). This could be used to detect local differences in material properties or magnetic anisotropy, for example, in addition to the tomography application.

Conventional x-ray tomography is enhanced through the use of multiple detectors by increasing signal acquisition rates and/or by combining information from a tilt series with information from difference in signal intensity, as described with respect to FIG. **5**, from the multiple detectors

Embodiments of the invention substantially increase X-ray count rates compared to most prior art detectors. Unlike the prior art, in which the sample needs to be tilted toward the x-ray detector, embodiments of the invention allow detection at any tilt angle of the specimen under observation or at any stage rotation, because the detectors surround the sample. In some embodiments, the functionality of the cold trap and cooling for the detectors is combined in one liquid nitrogen Dewar, rather than the two Dewars normally required, thereby reducing the chance of adverse effects on the image resolution of the microscope. The multiple detectors and the large solid detection angle, preferably about one steradian or greater, allows X-ray mapping, which previously required more than 1 hour measurement time, to be performed in a few minutes.

The invention also leads to new possibilities for X-ray detection, such as 3D X-ray tomography and depth determination of features in the specimen. Here the independent

directional detection of the X-rays received by the multiple detectors carry the information about the exact 3D position of the area on the sample emitting the X-rays.

While the embodiments described above describe the implementation of x-ray detectors for a transmission electron microscope, the invention is not limited to implementation ina TEM, but can be implemented in other instruments, such as scanning electron microscopes and scanning transmission electron microscopes.

A preferred method or apparatus of the present invention has many novel aspects, and because the invention can be embodied in different methods or apparatuses for different purposes, not every aspect need be present in every embodiment. Moreover, many of the aspects of the described embodiments may be separately patentable.

The drawings are intended to aid in understanding the present invention and, unless otherwise indicated, are not drawn to scale.

Although the present invention and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made to the embodiments described herein without departing from the spirit and scope of the invention as defined by the appended claims. Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present invention, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present invention. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

We claim as follows:

1. A method of determining three-dimensional positional and compositional information about a microscopic specimen, comprising:

scanning an electron beam across the sample;

detecting x-rays characteristic of the material in the sample using multiple x-ray angularly spaced around the sample, the x-rays caused to be emitted from the sample by the impact of the electron beam;

determining from the difference in the x-ray signals detected by the multiple detectors the position of the source of the x-rays along the beam direction; and

improving the accuracy of compositional information by means of the difference in attenuation of the x-ray signal detected by the multiple detectors.

2. The method of claim **1** in which determining from the difference in the x-ray signals detected by the multiple detectors the position of the source of the x-rays in the beam direction includes determining the position of the source of X-rays from the different attenuation of the x-ray signal caused by different path lengths through the sample of the x-rays detected by different detectors.

3. A method of determining three-dimensional positional and compositional information about a microscopic specimen, comprising:

scanning an electron beam across the sample;

detecting x-rays characteristic of the material in the sample using multiple x-ray angularly spaced around the sample, the x-rays caused to be emitted from the sample by the impact of the electron beam; and

determining from the difference in the x-ray signals detected by the multiple detectors the position of the source of the x-rays along the beam direction.

4. The method of claim **3** in which determining from the difference in the x-ray signals detected by the multiple detectors the position of the source of the x-rays in the beam direction includes determining the position of the source of X-rays from the different attenuation of the x-ray signal caused by different path lengths through the sample of the x-rays detected by different detectors.

5. A method of determining characteristics of a material, comprising:

directing an electron beam to impact the surface of a sample;

detecting x-rays caused to be emitted from the sample by the impact of the electron beam and characteristic of the material in the sample using at least two x-ray detectors angularly spaced around the sample and within the lens of a transmission electron microscope;

cooling a cold shield at least partly surrounding the detector; and

providing heat to maintain the detector at a higher temperature than the cold shield.

6. The method of claim **5** in which detecting x-rays using at least two x-ray detectors supported within the lens of a transmission electron microscope comprises detecting x-rays using at least two x-ray detectors supported by a pole piece of the transmission electron microscope.

7. The method of claim **5** further comprising maintaining a shutter moveable between an open position and a closed position for protecting the detector when it is not in use, at a temperature colder than that of the detector.

8. The method of claim **5** in which detecting x-rays using at least two x-ray detectors comprises detecting x-rays using at least two x-ray detectors positioned to receive x-rays having takeoff angles extending from less than 20 degrees to greater than 30 degrees.

9. The method of claim **5** in which detecting x-rays using at least two x-ray detectors comprises detecting x-rays using at least two SDD x-ray detectors, each SDD x-ray detector having an active area of greater than 10 mm^2.

10. The method of claim **5** in which detecting x-rays using at least two x-ray detectors comprises detecting x-rays using at least two x-ray detectors having active areas that together subtend a solid angle of greater than about 0.3 sr.

11. The method of claim **5** in which providing heat to maintain the detector at a higher temperature than the cold shield comprises providing heat from a resistive heater.

12. The method of claim **5** further comprising reducing the transmission of vibration between the detectors and the pole piece by mounting the detectors on a mount that includes a floating portion that supports the detectors.

13. The method of claim **12** in which the floating portion comprises multiple leaf springs.

14. The method of claim **5** further comprising maintaining a thermally resistive thermal connection between the cold shield and the detector.

15. The method of claim **5** in which cooling a cold shield at least partly surrounding the detector comprises providing a single cold source connected to the cold shield to cool the cold shield and the detector.

16. The method of claim **5** in which detecting x-rays using at least two x-ray detectors comprises detecting x-rays using four x-ray detectors.

17. The method of claim **5** further comprising reducing detection of x-rays that do not derive from the location of the electron beam impact with the sample using a collimator.

18. The method of claim **5** in which cooling the cold shield includes cooling a portion of the mount that comprises the cold shield.

19. The method of claim 5 further comprising reducing detection of x-rays that do not derive from the location of the electron beam impact using a portion of the mount as a collimator.

20. The method of claim 5 further comprising providing a thin window positioned in front of the detectors.

21. The method of claim 5 in which directing an electron beam includes directing an electron beam within a transmission electron microscope, a scanning transmission electron microscope, or a scanning electron microscope.

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UNITED STATES PATENT AND TRADEMARK OFFICE

CERTIFICATE OF CORRECTION

PATENT NO. : 8,410,439 B2
APPLICATION NO. : 13/312689
DATED : April 2, 2013
INVENTOR(S) : Von Harrach et al.

It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Title Page 1, (73), under "Assignee" heading please correct 'PEI Company, Hillsboro, OR (US)' to read, --FEI Company, Hillsboro, OR (US)--

Signed and Sealed this
Twenty-fourth Day of September, 2013



Teresa Stanek Rea
Deputy Director of the United States Patent and Trademark Office